KRAS RNA-seq: screening and cell-line omission analysis

October 11, 2026 | Follow-up to the October 10 group meeting

1. Main findings

Adding the post-test effect cutoff reduces the full-fit joint list from **5,442 to 2,020 genes**. Across eight paired cell-line omissions, **1,515** of these genes pass the joint screen in at least seven of the same eight rounds; **1,291** pass in every round. These are descriptive stability lists for discussion and subsequent external evaluation.

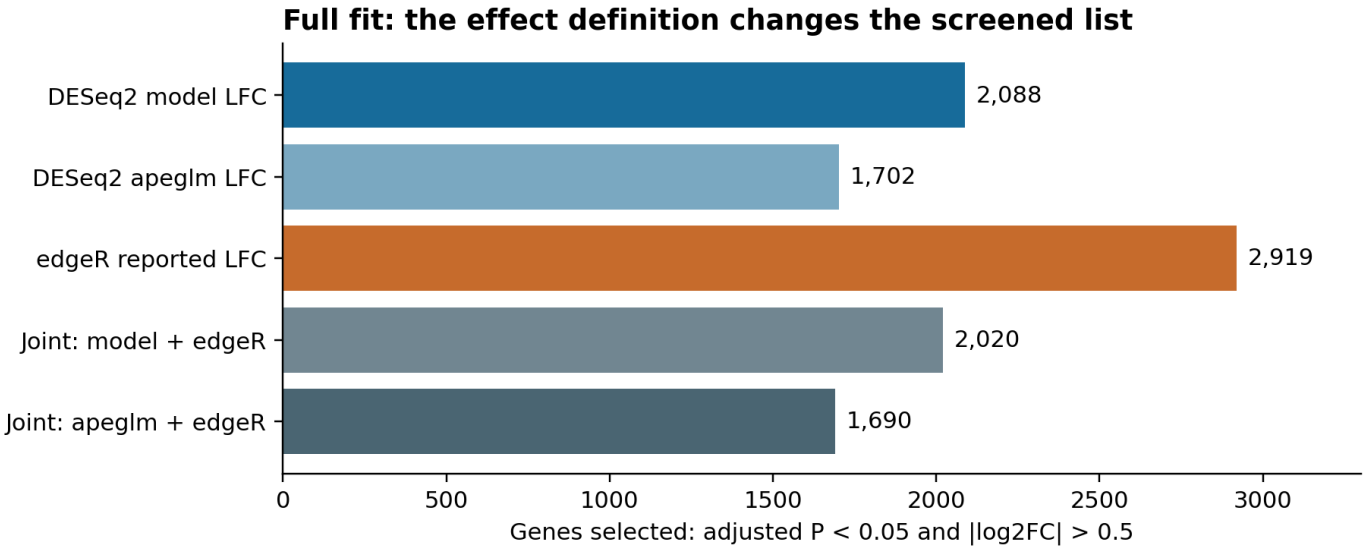


Figure 1. All counts refer to the full retained universe of 17,819 genes. The screen is adjusted P < 0.05 and |log2FC| > 0.5. DESeq2 model LFC is before apeglm shrinkage. edgeR uses its reported LFC with the default prior count 0.125. The 14,200-gene paper-comparison subset is not used here.

Definition	DESeq2	edgeR	Intersection
FDR < 0.05 only	5,943	6,431	5,442
FDR + primary effect cutoff	2,088	2,919	2,020
FDR + DESeq2 apeglm cutoff	1,702	2,919	1,690

Direction: 2,017/2,020 genes have finite effects with the full-fit direction in all eight rounds in both methods. Three genes are filtered out in one round. No direction reversal occurs among available estimates in this reference list.

Examples: KRAS, DUSP6, SPRY4, CCND1 and EMP2 pass in all rounds in both methods. MYC remains significant and negative in every round but crosses the 0.5 effect boundary twice.

2. Sample handling and model audit

The input has 28,395 genes and 17 runs. MiaPaca-2 KRAS libraries are summed gene by gene as raw integers before filtering and normalization, yielding 16 analysis observations. The pooled Treatment observation has 68,124,190 assigned counts; the Control observation is retained separately.

Local label	Run accession	BioSample	Public alias
Control P5	SRR24828470	SAMN35622211	MP2_SC
KRAS P5	SRR24828471	SAMN35622210	MP2_K2v2
KRAS P5_v2	SRR24828472	SAMN35622209	MP2_K2

The local v2 label is reversed relative to the public alias. The archived ENA snapshot (2026-10-04) confirms distinct BioSample records and FASTQ MD5 values. It does not establish technical or biological repeat identity or shared biological material. Pooling remains an explicit exploratory assumption; the two libraries are not counted as independent replicates.

Audit item	Full fit	Each omission fit
Observations / cell lines	16 / 8	14 / 7
Model	~ condition + cell_line	~ condition + cell_line
Design rank / residual df	9 / 7	8 / 6
Low-count rule	>=10 counts in >=2 observations	Same rule, recalculated
Condition direction	KRAS siRNA / Control	KRAS siRNA / Control

Each omission removes both conditions of one cell line. Omitting MiaPaca-2 removes its Control and its pooled Treatment, including both source KRAS libraries. Normalization and dispersions are refitted for each reduced dataset. The additive model estimates a common cell-line-adjusted treatment effect without an interaction; within-cell-line replication is unavailable.

Primary effect and test are separate

DESeq2 uses Wald tests and its model log2FoldChange before apeglm shrinkage for the main screen. apeglm posterior effects and posterior SD are saved separately. edgeR uses TMM and robust quasi-likelihood fitting (v4, legacy = FALSE); its reported logFC is moderated by prior.count = 0.125. Both tests use a zero-effect null. The 0.5 cutoff is applied after testing.

Archived single-library checks

P5 choice	DESeq2 screen	edgeR screen	Intersection
Sum two libraries	2,088	2,919	2,020
Retain SRR24828472	2,094	2,827	2,008
Retain SRR24828471	2,086	2,855	1,991

These screens were recalculated from the archived Week 4 fits. Single-library models were not refitted in this work.

3. Effect stability across eight omissions

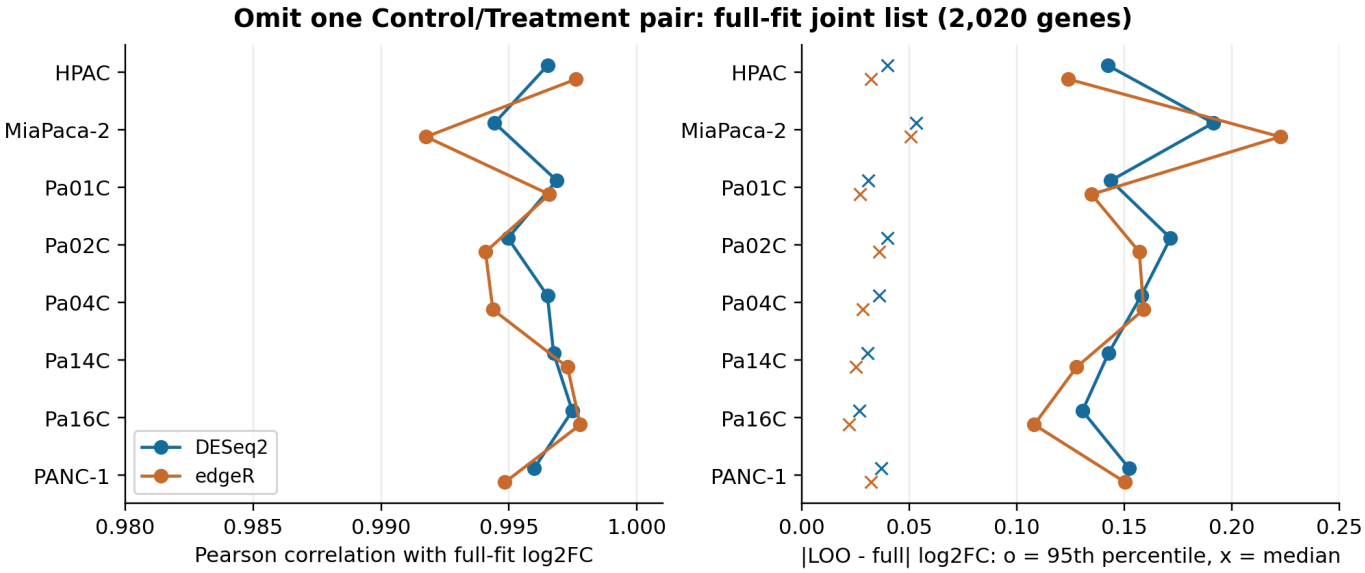


Figure 2. Effect comparisons use the full-fit joint list, with finite values in each round (2,020, except 2,018 after omitting PANC-1 and 2,019 after omitting Pa02C). Point estimates are compared with the full fit; these values are not confidence intervals.

DESeq2 Pearson correlations are 0.9944-0.9975 and edgeR correlations are 0.9918-0.9978. The 95th percentiles of absolute log2FC change are 0.1306-0.1915 and 0.1082-0.2226, respectively. Omitting MiaPaca-2 gives the lowest correlations and largest 95th-percentile changes in both methods, while overall agreement remains high.

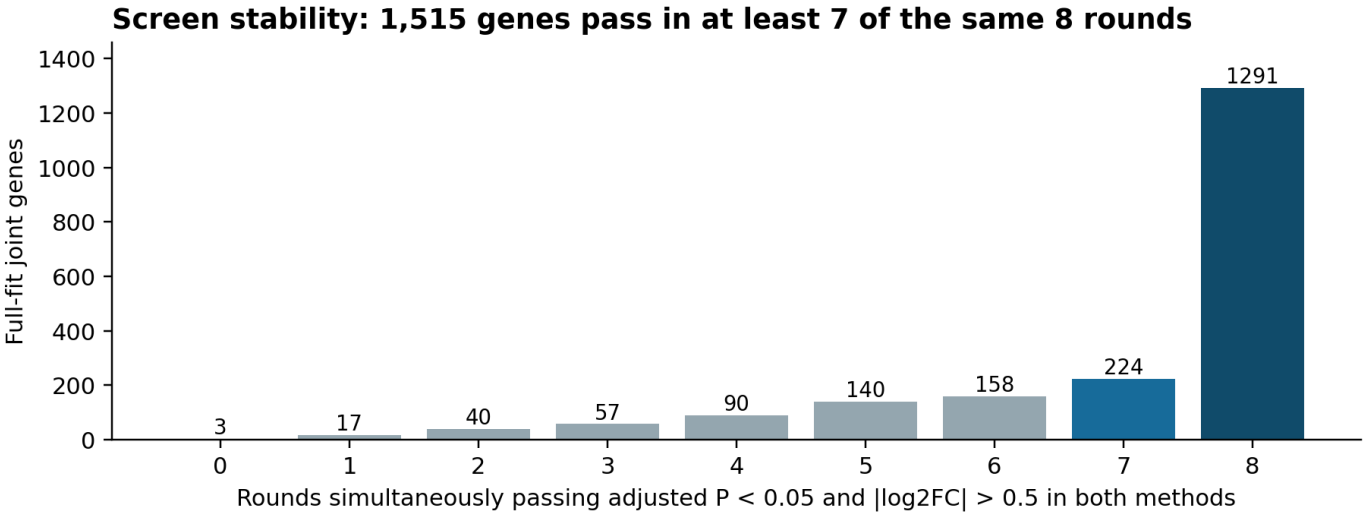


Figure 3. Joint passes require both methods to pass in the same round. The main descriptive list also requires full-fit selection, eight finite effects and direction agreement in each method. Separate >=7/8 passes per method would give 1,517 genes because the failed rounds can differ; the simultaneous definition gives 1,515.

GSTM2 and CDK15 lack estimates after omitting PANC-1; DDX3Y lacks estimates after omitting Pa02C. All three fail that round's low-count filter. Missing estimates are recorded explicitly. A total of 52 reference genes have a maximum effect change >=0.5 log2 units in at least one method and are exported for inspection; this flag does not establish single-line causation.

4. Candidate results for discussion

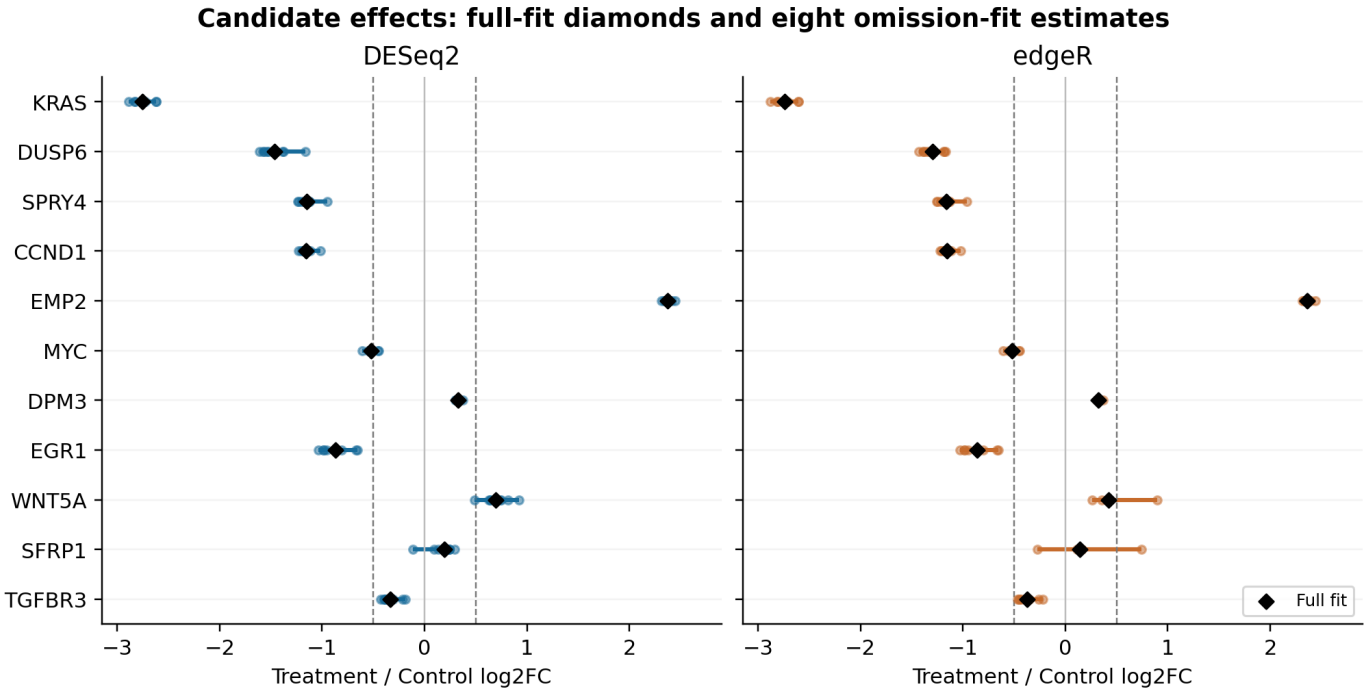


Figure 4. Diamonds show full-fit estimates; dots and horizontal ranges show the eight omission estimates. Dashed lines mark +/-0.5. Ranges are minimum-to-maximum point estimates, not confidence intervals.

Gene	DESeq2 full LFC	edgeR full LFC	Screen rounds D / E	Interpretation
KRAS	-2.755	-2.737	8/8 ; 8/8	Persistent decrease
DUSP6	-1.463	-1.298	8/8 ; 8/8	Persistent decrease
SPRY4	-1.148	-1.163	8/8 ; 8/8	Persistent decrease
CCND1	-1.155	-1.154	8/8 ; 8/8	Persistent decrease
EMP2	2.371	2.362	8/8 ; 8/8	Persistent increase
MYC	-0.519	-0.519	6/8 ; 6/8	Effect boundary sensitivity
DPM3	0.325	0.325	0/8 ; 0/8	Small effect despite FDR
EGR1	-0.869	-0.863	4/8 ; 1/8	Statistical evidence differs
WNT5A	0.694	0.420	0/8 ; 1/8	Method-sensitive evidence
SFRP1	0.196	0.141	0/8 ; 1/8	Direction changes
TGFBR3	-0.336	-0.376	0/8 ; 0/8	Below the effect cutoff

MYC remains FDR-significant and negative in all rounds; omitting HPAC or Pa16C moves its absolute effect below 0.5. DPM3 remains significant but never exceeds 0.5. EGR1 retains negative effects greater than 0.5 in magnitude but passes FDR in 4/8 DESeq2 and 1/8 edgeR fits. These are different forms of sensitivity, not interchangeable evidence of a dominant cell line.

5. Manuscript Methods draft

Count data and sample structure

We analysed the corrected gene-level count matrix retained in the project repository. The matrix contained 28,395 Entrez gene identifiers and 17 RNA-seq runs from eight pancreatic cancer cell lines: HPAC, MiaPaca-2, Pa01C, Pa02C, Pa04C, Pa14C, Pa16C and PANC-1. Each cell line had a control siRNA condition and a KRAS siRNA condition; MiaPaca-2 had two KRAS-condition libraries. We matched count columns to sample metadata, checked that counts were nonnegative integers, and compared every column with its source featureCounts table and assigned-count total. Gene annotations were matched by Entrez identifier. The corrected KRAS M1 counts were retained. This analysis began from the existing count tables; alignment and read quantification were not rerun in this work.

MiaPaca-2 library handling

Following the October 10 group discussion, we summed the two MiaPaca-2 KRAS-condition libraries gene by gene at the raw integer-count stage, before gene filtering or normalization. Local KRAS P5 corresponds to SRR24828471, BioSample SAMN35622210, public alias MP2_K2v2; local KRAS P5_v2 corresponds to SRR24828472, BioSample SAMN35622209, public alias MP2_K2. Their summed assigned-count total was 68,124,190. The control library was SRR24828470, BioSample SAMN35622211, alias MP2_SC. The archived ENA snapshot dated October 4, 2026 records different BioSample accessions and FASTQ MD5 values for the two KRAS libraries. These records do not establish whether they represent technical or biological repeats or the same biological material. We therefore treated the summed counts as one analysis observation under an explicit exploratory pooling assumption, rather than counting the two libraries as independent replicates. The original run-level tables were retained, together with the previous analyses that used either library separately.

Differential expression

The full analysis contained 16 observations, comprising one control and one treatment observation for each cell line. Genes were retained if they had at least 10 counts in at least two observations. We fitted the additive design ~ `condition + cell_line`, with Control as the reference level and Treatment representing KRAS siRNA. The treatment coefficient estimated a common cell-line-adjusted effect. Positive log2 fold change indicates higher expression after KRAS siRNA. The full design had rank nine and seven residual degrees of freedom. It did not include a treatment-by-cell-line interaction, and the available design did not provide within-cell-line replication.

Analyses used R 4.5.2, DESeq2 1.50.2, edgeR 4.8.2 and apeglm 1.32.0. DESeq2 used median-of-ratios size factors, negative-binomial dispersion estimation and Wald tests, with independent filtering at `alpha = 0.05`. Its primary effect column, `log2FoldChange`, was the model estimate before apeglm shrinkage. We separately calculated apeglm posterior effect estimates and posterior standard deviations; shrinkage did not supply the primary Wald-test P values. edgeR used TMM normalization, robust dispersion estimation, robust quasi-likelihood fitting and a quasi-likelihood F test for the treatment coefficient, followed by Benjamini-Hochberg adjustment. We retained edgeR v4's default `legacy = FALSE` fitting behaviour. The reported edgeR `logFC` used its default prior count of 0.125 for effect estimation; it was not an apeglm estimate. Reverse contrasts were checked to negate the reported effect and preserve the raw P value in both methods.

We selected genes after testing using adjusted $P < 0.05$ and absolute log2 fold change > 0.5. DESeq2 used its estimate before apeglm shrinkage; edgeR used its reported `logFC`. This selection did not change the zero-effect null hypothesis into an effect-threshold test. A separate audit applied the same selection to DESeq2 apeglm effects to show the consequence of that alternative definition. Counts and overlaps were calculated on the full retained gene universe, rather than the 14,200-gene subset previously used for comparison with the published study.

6. Methods: omission analysis and verification

Leave-one-cell-line-out analysis

We performed eight additional fits. Each fit removed both the Control and Treatment observations for one cell line. Omitting MiaPaca-2 also removed the pooled treatment observation containing both KRAS libraries. Each reduced dataset contained 14 observations from seven cell lines. Unused factor levels were dropped; every design had rank eight and six residual degrees of freedom. We reapplied the same count filter within each reduced dataset and re-estimated normalization factors, dispersions, effects and tests with the same model and software settings. Genes excluded by filtering were recorded as unavailable rather than assigned zero effects or P values.

We compared each omission fit with the full fit using effect correlations, direction agreement, absolute effect changes, and overlap of the post-test screened sets. We tabulated each gene's effect range, number of finite estimates, number of available adjusted P values, direction agreement and screen-pass count. Full-fit screened sets were fixed reference lists for the stability assessment. The descriptive joint list required full-fit selection by both methods, finite estimates in all eight omission fits in both methods, the full-fit direction in every fit, and simultaneous selection by both methods in at least seven of the same eight rounds. We also reported the stricter eight-of-eight list. These correlated fits were sensitivity analyses; their P values were not pooled or averaged, and no additional error-control guarantee was assigned to the descriptive stability lists. A maximum absolute change of at least 0.5 log2 units was flagged for inspection, without treating this display rule as evidence of single-cell-line causation. No pathway enrichment, patient-outcome validation or new experimental work was performed in this analysis.

Completed verification

Raw counts were checked against all 17 source tables, assigned totals, metadata order and the saved ENA alias mapping. The pooled counts match exact gene-level sums. All nine designs are full rank; reverse contrasts negate effects and preserve raw P values; no DESeq2 coefficient fit failed to converge. The full fit matches the archived Week 4 effect estimates and P values to numerical tolerance. Two fresh nine-fit runs produced 68 byte-identical TSV and method-JSON files.

Limits and next discussion

The descriptive 7/8 and 8/8 lists have not been evaluated for patient-outcome prediction. No new guarantee of false-discovery control follows from counting correlated omission passes. P5 repeat identity remains unresolved. Effect changes do not establish direct regulation, protein changes or a validated clinical signature. The existing Week 4 paired-expression appendix is retained; it was not regenerated.

The next meeting can discuss the omission definition, candidate effects and the joint list. The October 10 meeting suggested Lin draft Methods and other group members draft Results, with roles open to group adjustment. The manuscript draft was requested within two weeks of that meeting; no exact delivery time or fixed word limit was assigned. Pathway enrichment remains deferred.

Deliverables and references

The dated analysis directory contains code, full tables for nine fits, gene and candidate stability tables, screening audits, four figures, Methods_en.md and Results_zh.md. Input files and history remain in the independent local repository branch. No remote push or publication was performed.

Source records: corrected project counts (analysis/2026-10-01/counts); archived Week 4 results and ENA snapshot; local October 10 meeting summary. Statistical software: DESeq2 (doi:10.1186/s13059-014-0550-8), edgeR v4 (doi:10.1093/nar/gkaf018), apeglm (doi:10.1093/bioinformatics/bty895). These references identify the software; the Methods describe the operations actually executed here.